



L4b: Single Asset Geometric Brownian Motion Models

CHEME 5660 Cornell University

Copyright Jeffrey Varner 2026. All Rights Reserved

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we introduce our first continuous-time price model, geometric Brownian motion, estimate its parameters from data, test it against the stylized facts, and compute a closed-form target probability for a scheduled trade.

Today's concepts

- **Model share prices with GBM**: the stochastic differential equation, the Wiener process, the exact lognormal solution, and the exact one-step transition used for Monte Carlo.
- **Estimate drift and volatility**: derive the growth-rate law, estimate the mean growth rate and volatility from a growth-rate series, and recover the arithmetic drift.
- **Test the model and evaluate a trade**: check GBM against the stylized facts, then derive the probability that a scheduled long position clears a target return.

Lectures and Examples

Lecture:

CHEME-5660-L4b-Lecture-SingleAsset-GeometricBrownianMotion-TradeRule-Fall-2026.ipynb

Example 1:

CHEME-5660-L4b-Example-Parameters-SAGBM-Fall-2026.ipynb

Example 2:

CHEME-5660-L4b-Example-GBM-NPV-TradeRule-Fall-2026.ipynb

The first example estimates the mean growth rate and volatility of a GBM model from historical prices, recovers the drift, and simulates sample paths. The second computes the probability that a scheduled long position clears a target present-value return.

Company Profile: Jane Street

Jane Street is a privately held **quantitative trading firm**: it trades across asset classes on more than 200 venues, deploys mainly its own capital, and is a major market maker in exchange-traded funds.

- **Client posture**: Citadel Securities handles a large share of U.S. retail order flow; Jane Street is primarily proprietary and institution facing.
- **Product emphasis**: Jane Street is synonymous with ETFs; Citadel Securities with U.S. equities, options, and FX market making.
- **Engineering culture**: Jane Street optimizes for correctness and composability (functional programming in OCaml); Citadel Securities for reliability and throughput at scale.

Role in this lecture: both firms depend on quantitative models of prices and execution; today we build one baseline price model.

Concept Review: N-Ary Tree Models

A binomial lattice allows only two price movements per step. If two possible futures are not enough, an **N-ary lattice** allows m branches with factors $f_1, \dots, f_m$ and probabilities $p_1, \dots, p_m$.

- A recombining node is indexed by its branch-count vector; its probability is multinomial (L4a).
- More branches make the set of possible prices finer (at the cost of more states); a smaller time step brings the steps closer together.

Today's idea: replace the discrete set of prices with a **continuous** one and the discrete steps with continuous time. Refinement alone is not enough; with the right scaling the lattice converges to a continuous model, and that model is geometric Brownian motion.

Example: [CHEME-5660-L4a-Example-N-Ary-Lattice-Fall-2026.ipynb](#)

Single Asset Geometric Brownian Motion (GBM)

Let $S(t) > 0$ be the share price at time t on a horizon $0 \leq t \leq T$. Let $\mu \in \mathbb{R}$ (units: inverse years) be a constant arithmetic drift, $\sigma > 0$ (units: inverse years to the one-half power) a constant **volatility**, and $dW(t)$ the increment of a Wiener process (next frame). Geometric Brownian motion (GBM) is a continuous-time random walk whose drift and noise are both proportional to the price:

$$\frac{dS(t)}{S(t)} = \underbrace{\mu dt}_{\text{drift}} + \underbrace{\sigma dW(t)}_{\text{noise}}.$$

We will see that the log price is a Brownian motion with drift. We reserve g for an observed continuously compounded growth rate. The drift μ and a growth rate g share inverse-time units, but they are different quantities; the relationship is made precise below.

The Wiener Process

A Wiener process $\{W(t), 0 \leq t \leq T\}$ is a continuous real-valued stochastic process with three properties:

- The noise vanishes at zero: $W(0) = 0$ with probability one.
- Increments over disjoint intervals $W(t_1) - W(t_0), \dots, W(t_k) - W(t_{k-1})$ are independent for any $0 \leq t_0 < t_1 < \dots < t_k \leq T$.
- Each increment is normal with mean zero and variance equal to the elapsed time, $W(t) - W(s) \sim \mathcal{N}(0, t - s)$ for $0 \leq s < t \leq T$.

The third property lets us write any increment as a scaled standard normal:

$$W(t) - W(s) = \sqrt{t - s} Z, \quad Z \sim \mathcal{N}(0, 1).$$

The Analytical Solution

Take $t_0 = 0$ and let S_0 be today's price. Solving the SDE with Itô calculus gives the share price at a fixed future time T :

$$S_T = S_0 \exp \left[\underbrace{(\mu - \frac{\sigma^2}{2})T}_{\text{mean log growth}} + \underbrace{\sigma\sqrt{T}Z}_{\text{shock}} \right], \quad Z \sim \mathcal{N}(0, 1).$$

Thus the log price ratio $\ln(S_T/S_0)$ is normal with mean $(\mu - \sigma^2/2)T$ and variance σ^2T , and the price S_T is **lognormal** and strictly positive.

The $-\sigma^2/2$ term comes out of the Itô calculus in the derivation notebook; it is the **half-variance correction** we meet again below.

Derivation: [CHEME-5660-L4b-GBM-Solution-Derivation-Fall-2026.ipynb](#)

Mean and Variance of the GBM Price

From the lognormal moments, the expectation and variance of S_T are:

$$\underbrace{\mathbb{E}(S_T)}_{\text{mean price}} = S_0 e^{\mu T}, \quad \underbrace{\text{Var}(S_T)}_{\text{price variance}} = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1).$$

The expected price grows at the arithmetic drift μ . The median price $S_0 e^{(\mu - \sigma^2/2)T}$ grows at the smaller rate $\mu - \sigma^2/2$.

The gap is the **half-variance correction**: the lognormal distribution is right skewed, so its mean sits above its median.

Discrete Time GBM Model

We observe prices on a grid $t_j = j\Delta t$, $j = 0, 1, \dots, N$, with time step Δt (years) and horizon $T = N\Delta t$. Applying the fixed-horizon solution over one step gives the **one-step transition**:

$$S_{t_j} = S_{t_{j-1}} \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right], \quad j = 1, \dots, N,$$

with $Z_1, \dots, Z_N$ **independent** standard normals. For constant μ and σ this is exact at the grid points, not a numerical time-stepping approximation.

Jumping directly from S_0 to t_j uses the cumulative shock $\tilde{Z}_j = j^{-1/2} \sum_{i \leq j} Z_i$:

$$S_{t_j} = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)t_j + \sigma\sqrt{t_j} \tilde{Z}_j\right].$$

$\tilde{Z}_j$ is standard normal at each j but **not** independent across j (correlation $\sqrt{t_j/t_k}$ for $j < k$): use it for one horizon, use the transition for paths.

Monte Carlo Simulation of GBM

Given S_0 , μ , σ , Δt , N steps, and M paths, store the simulated prices in $\mathbf{S} \in \mathbb{R}^{(N+1) \times M}$, one path per column, rows indexed by $j = 0, \dots, N$.

For each sample path (each column of $\mathbf{S}$):

1. Set the price in row $j = 0$ to S_0 .
2. For $j = 1, \dots, N$: draw $Z_j \sim \mathcal{N}(0, 1)$ and apply the one-step transition:

$$S_{t_j} \leftarrow S_{t_{j-1}} \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t} Z_j\right].$$

The result is M possible future trajectories of N steps each. Their spread is process uncertainty; the uncertainty in μ and σ is a separate layer.

Implementation: the `sample(...)` function in the course package.

Estimation of GBM Parameters

Divide the one-step transition by $S_{t_{j-1}}$, take logarithms, and divide by Δt . The **one-step growth rate** g_j over $t_{j-1} \rightarrow t_j$, the same quantity we computed from data in L3a, is:

$$g_j \equiv \frac{1}{\Delta t} \ln \frac{S_{t_j}}{S_{t_{j-1}}} = \underbrace{\left(\mu - \frac{\sigma^2}{2}\right)}_{\text{mean growth } \bar{g}} + \underbrace{\frac{\sigma}{\sqrt{\Delta t}}}_{\text{growth-rate std}} Z_j.$$

Because the Z_j are independent standard normals, growth rates over equally spaced, non-overlapping steps are independent and normal with mean $\bar{g} \equiv \mu - \sigma^2/2$ and variance $\sigma^2/\Delta t$.

- **Drift versus growth:** μ is the SDE drift; the mean growth rate $\bar{g}$ is what a growth-rate series measures. They differ by the half-variance correction.
- **Risk:** σ sets the growth-rate standard deviation $\sigma/\sqrt{\Delta t}$ and the one-step log-return standard deviation $\sigma\sqrt{\Delta t}$.

Volatility

From $N + 1$ prices on the grid we form N growth rates $g_1, \dots, g_N$. Their sample mean g' and sample standard deviation σ_g (our L3a risk measure) are:

$$g' = \frac{1}{N} \sum_{j=1}^N g_j, \quad \underbrace{\sigma_g}_{\text{growth-rate std}} = \sqrt{\frac{1}{N-1} \sum_{j=1}^N (g_j - g')^2}.$$

(Denominator $N - 1$: N observations less the one mean estimated from them.) But σ_g is **not** the GBM volatility. Since $\text{Var}(g_j) = \sigma^2 / \Delta t$, σ_g estimates $\sigma / \sqrt{\Delta t}$, and the volatility estimate is:

$$\hat{\sigma} = \sigma_g \sqrt{\Delta t}.$$

For daily data $\Delta t = 1/252$: divide the daily growth-rate standard deviation by $\sqrt{252}$, or equivalently multiply the daily log-return standard deviation by $\sqrt{252}$.

Drift

Under GBM the arithmetic drift is $\mu = \bar{g} + \sigma^2/2$. Given an estimate $\hat{g}$ of the mean growth rate, the drift estimate is:

$$\underbrace{\hat{\mu}}_{\text{SDE drift}} = \underbrace{\hat{g}}_{\text{mean growth}} + \underbrace{\frac{1}{2}\hat{\sigma}^2}_{\text{correction}}.$$

Two common ways to estimate $\bar{g}$:

- **Method 1, average growth rate:** $\hat{g} = g'$, the sample mean, which is also the maximum likelihood estimate under GBM.
- **Method 2, linear regression:** regress log price on time; the slope estimates $\bar{g}$, but the residuals are a Brownian path, so the ordinary standard errors and residual checks do not apply.

The drift is the harder estimate: mean growth is small relative to growth-rate noise, so both methods carry substantial sampling error.

Linear Regression: The Model

Take the logarithm of the cumulative GBM solution at grid point t_j :

$$\ln S_{t_j} = \underbrace{\ln S_0}_{\text{intercept}} + \underbrace{\bar{g} t_j}_{\text{trend}} + \underbrace{\sigma W(t_j)}_{\text{error}}, \quad j = 0, \dots, N.$$

The noise has mean zero, so the **expected** log price is a line with slope $\bar{g}$: regressing log price on time recovers that line, and the Wiener term is the error term.

That error is a Brownian path: neighboring values are strongly correlated, its variance $\sigma^2 t_j$ grows along the sample, and the fitted residuals inherit both features. Least squares still gives a sensible slope, but textbook standard errors, confidence intervals, and residual-normality tests assume independent, identically distributed errors and are not calibrated here.

Example 1: the Anderson–Darling normality test on these residuals rejects: a warning about the test's assumptions, not a verdict on GBM.

Linear Regression: The Normal Equations

Stack the $N + 1$ log prices as an overdetermined system $\hat{\mathbf{X}}\theta + \epsilon = \mathbf{y}$, where θ holds the intercept $\ln S_0$ (left free, not pinned to day one) and the slope $\bar{g}$, ϵ is the error vector, and the augmented design matrix $\hat{\mathbf{X}}$ carries a column of ones:

$$\hat{\mathbf{X}} = \begin{bmatrix} 1 & t_0 \\ 1 & t_1 \\ \vdots & \vdots \\ 1 & t_N \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} \ln S_{t_0} \\ \ln S_{t_1} \\ \vdots \\ \ln S_{t_N} \end{bmatrix}.$$

The least-squares estimate is the solution of the normal equations:

$$\boxed{\hat{\theta} = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \mathbf{y}.}$$

Its second component is the regression estimate $\hat{g}$.

Example 1: CHEME-5660-L4b-Example-Parameters-SAGBM-Fall-2026.ipynb

Stylized Facts

Does GBM capture the empirical regularities of asset growth rates?

- **Heavy tails:** empirical growth rates have heavier tails than a normal distribution, so extreme moves are more likely than a Gaussian predicts.
- **Absence of autocorrelation:** if prices follow a random walk, growth rates over non-overlapping intervals are uncorrelated, and there is no predictability to exploit.
- **Volatility clustering:** large moves tend to follow large moves and small moves follow small moves, so volatility alternates between high and low regimes.

We test each against the growth-rate law we just derived.

Growth Over a Horizon Under GBM

Define the **horizon-average growth rate** over $0 \rightarrow T$ as $g_{T,0} \equiv (1/T) \ln(S_T/S_0)$, in the same to-from subscript order as the discount factor $\mathcal{D}_{T,0}$. Substituting the GBM solution and dividing by T gives:

$$\underbrace{g_{T,0}}_{\text{horizon-average growth}} = \bar{g} + \frac{\sigma}{\sqrt{T}} Z, \quad Z \sim \mathcal{N}(0, 1).$$

So $g_{T,0}$ is normal with mean $\bar{g}$ and variance σ^2/T ; the one-step case is $T = \Delta t$.

The uncertainty of the horizon-average growth rate shrinks as T grows while its mean stays put. The price does not settle down: $\ln(S_T/S_0) = T g_{T,0}$ has variance $\sigma^2 T$, which grows with T .

Heavy Tails and Volatility Clustering

Heavy tails: not captured. One-step and horizon-average growth rates are both normal under GBM. The lognormal price has a long right tail, but that is not a heavy-tailed **growth-rate** distribution.

Volatility clustering: not captured. The volatility σ is constant, so there is no mechanism for high- and low-volatility periods to alternate.

Both failures follow from the same assumption: independent Gaussian increments with a constant variance rate. That assumption is what makes the model exactly solvable.

Autocorrelation of One-Step Growth Rates

From the growth-rate law, $g_j = \bar{g} + (\sigma/\sqrt{\Delta t})Z_j$ with independent shocks. The autocovariance at lag $\tau \geq 1$ is:

$$\text{Cov}(g_j, g_{j+\tau}) = \frac{\sigma^2}{\Delta t} \text{Cov}(Z_j, Z_{j+\tau}) = 0,$$

because Z_j and $Z_{j+\tau}$ are independent. With $\text{Var}(g_j) = \sigma^2/\Delta t$, the autocorrelation function is:

$$\rho_g(\tau) = \frac{\text{Cov}(g_j, g_{j+\tau})}{\text{Var}(g_j)} = 0, \quad \tau \geq 1.$$

Absence of autocorrelation: captured for one-step growth rates over non-overlapping intervals. This says nothing about squared growth rates, whose dependence (L3a) GBM also cannot produce.

GBM Trade Rule

Buy $n_0 > 0$ shares at S_0 at time 0; sell all n_0 shares at S_T at the scheduled exit $T = N\Delta t$, $N \geq 1$ (frictionless: no costs, price return only). With annual continuously compounded benchmark growth rate g_b , $\mathcal{D}_{T,0}(g_b) = e^{g_b T}$ is the accumulation factor and $\mathcal{D}_{T,0}^{-1}(g_b) = e^{-g_b T}$ discounts the sale to time 0:

$$\text{NPV}(g_b, T) = \underbrace{-n_0 S_0}_{\text{entry (now)}} + \underbrace{n_0 S_T \mathcal{D}_{T,0}^{-1}(g_b)}_{\text{exit (future)}}.$$

Dividing by the initial investment gives the **scaled NPV**, a dimensionless present-value return on the share cost:

$$\underbrace{\rho_T}_{\text{scaled NPV}} := \frac{\text{NPV}(g_b, T)}{n_0 S_0} = \left(\frac{S_T}{S_0} \right) e^{-g_b T} - 1.$$

This is the L4a rule with one change: S_T now follows GBM instead of a binomial lattice.

Substitute the GBM Solution

Replacing S_T by the GBM solution puts all of the randomness in one standard normal Z :

$$\rho_T = \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}Z\right] \underbrace{e^{-g_b T}}_{\text{discount}} - 1, \quad Z \sim \mathcal{N}(0, 1).$$

- **Short holding period:** for a few days or weeks and a realistic benchmark, $|g_b|T$ is small, $e^{-g_b T} \approx 1$, and $\rho_T \approx S_T/S_0 - 1$ is the fractional price change.
- **Longer holding period:** whenever $|g_b|T$ is not small, keep the discount factor.

Either way ρ_T is a continuous random variable driven by Z , so a target probability is a one-line standardization.

Cumulative Probability of a Trade Exit

Let $\rho_\star > -1$ be a target (under GBM $S_T > 0$, so $\rho_T > -1$ always). We ask for the probability that ρ_T at the scheduled exit **strictly exceeds** the target. Moving the non-random terms to the right:

$$\mathbb{P}(\rho_T > \rho_\star) = \mathbb{P}\left(\ln \frac{S_T}{S_0} > \ln(1 + \rho_\star) + g_b T\right).$$

Under GBM, $\ln(S_T/S_0)$ is normal with mean $(\mu - \sigma^2/2)T$ and standard deviation $\sigma\sqrt{T}$. Standardizing both sides:

$$\mathbb{P}(\rho_T > \rho_\star) = \mathbb{P}\left(\underbrace{\frac{\ln(S_T/S_0) - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}}}_{Z \sim \mathcal{N}(0,1)} > \frac{\ln(1 + \rho_\star) + g_b T - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}}\right).$$

As in L4a, this is the scheduled terminal exit; a monitored take-profit or stop-loss is a first-passage question.

The Closed-Form Target Probability

A standard normal exceeds z with probability $1 - \Phi(z)$, where Φ is the standard normal cumulative distribution function. Therefore:

$$\mathbb{P}(\rho_T > \rho_\star) = 1 - \Phi\left(\frac{\ln(1 + \rho_\star) + g_b T - (\mu - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right).$$

- The probability uses the **real-world** μ and σ estimated from data. A risk-neutral drift, which we will use to price derivatives later, changes the measure and no longer answers a probability-of-profit question.
- For a short holding period $g_b T \approx 0$ and the benchmark drops out of the numerator.
- A target of -1 or below is cleared with probability one.

Example 2: CHEME-5660-L4b-Example-GBM-NPV-TradeRule-Fall-2026.ipynb

Optional Advanced Material

Lattice to GBM advanced/lattice-limit/

Match the binomial factors to μ and σ and watch the terminal lattice distribution converge to the lognormal as the step count grows.

First-passage rules under GBM

advanced/first-passage/

A closed form for reaching one barrier, the L4a recursion and Monte Carlo for two, and the effect of the monitoring frequency.

Drift uncertainty

advanced/drift-uncertainty/

Why the drift estimate depends on the calendar span, not the sampling frequency, and what that does to the target probability.

Monte Carlo versus analytic

advanced/monte-carlo/

Estimate the target probability by simulation with standard errors, and compare the exact transition with an Euler scheme.

Notebooks: [lectures/week-4/L4b/advanced](#), also linked from the lecture notebook.

Summary

Today we introduced geometric Brownian motion as our first continuous-time price model, estimated its parameters from growth-rate data, tested it against the stylized facts, and derived a closed-form probability for a scheduled trade.

Key takeaways

- **GBM is a tractable lognormal model:** exact solution, exact one-step simulation with independent normal shocks, mean growing at the drift.
- **Drift and growth are different parameters:** the growth-rate series gives $\bar{g}$ and, through $\sigma_g \sqrt{\Delta t}$, the volatility; the drift follows from the half-variance correction.
- **One stylized fact passed, two failed:** growth rates are uncorrelated but Gaussian with constant volatility; the terminal target probability is one minus a standard normal CDF.

Next, we extend GBM to several correlated assets and use it to build portfolios.

Today: the lattice becomes a continuous process, and the trade rule gets a closed form.